

orbit_demo design overview

Package section 01_overview of orbit_demo_design_review, written 2026-10-06 from the package files and the repository at commit 3dffffbb. Links are relative to this file. Cited reports/... files are linked to their byte-identical copies in 07_verification. Unlinked repository paths under rtl/, pd/, tb/, fault/ and scripts/ are copied in 04_digital_design/source/. Only build/... files without a package copy (the copies are listed in README.md under How the package was built), docs/research/... and ORFS image paths (platforms/..., flow/scripts) are outside the package. Discrepancies are listed in the discrepancy register (09_review_notes).

Status labels: **VERIFIED** = a tool log or report shows it. **REPRODUCED 2026-10-06** = re-run for this package. **TARGET** = design goal, not measured. **ASSUMPTION. CLAIM (unverified)** = stated in a document, no log found. **MISSING. N/A** = not applicable (reason given).

Design revision covered by this package

Item	Revision / identifier	Note
Design	orbit_demo (ORBIT-AI 4-lane INT8 digital demonstrator)	top module orbit_demo, LANES=4
RTL	rtl/orbit_demo.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_keep_reg.v at git commit 26d71a8 (originally committed 2026-09-29 18:54 UTC; see the note on commit dates below); unchanged through the package commit	sha256 a3fffb22... / efef3d33... / bba77a5a... / ea05bdc1... (full hashes in MANIFEST.sha256 at the package root)
Specification	docs/SPEC.md at 26d71a8 (never revised)	sha256 aef5d9fb...
Concept brief	docs/orbit-ai-design-brief.pdf, "ORBIT-AI v0.1, 29 September 2026" (never revised; describes an earlier project state, see discrepancy register)	sha256 9d5f33d3...
Reviewed layout	ORFS sky130hd, variant sep (copy-separation fences), run 2026-09-29 21:56-22:12 UTC, outputs build/pd_sep/results/sky130hd/orbit_demo/sep/	6_final.gds sha256 cd19afa9...; 6_final.def f5c544f3...; 6_final.v 68093c41...; 6_final.spef 014e655a...
Layout configuration	pd/sky130hd_sep/config.mk and regions.tcl at 5e54769; constraint.sdc at 3e8dad1 (set clk_period 7.2)	The run (21:56 UTC) used config.mk/regions.tcl as in d8ead98, identical to 5e54769 once comments are stripped. It used the working-tree constraint.sdc with period 7.2 ns, which is the version in commit 3e8dad1 (originally committed at 22:00 UTC, after the run had started) and is byte-identical to the flow copy build/pd_sep/sdc/sky130hd/sep.sdc (written 21:56 UTC). d8ead98's constraint.sdc has a 7.0 ns period, so a rebuild must not use it
Flow / tools	ORFS docker image tag openroad/orfs:latest; the only local image is sha256:2e5bf6fe865e... (created 2026-09-29 02:00 UTC, before the run). Linking the run to this digest is an ASSUMPTION: the run logged no digest. OpenROAD prints version "unknown". KLayout 0.30.12 (DRC/LVS). Yosys 0.69+154 (local sims/schematics)	No PDK commit is recorded in the logs
Process / library	SkyWater SKY130, sky130_fd_sc_hd, liberty sky130_fd_sc_hd_tt_025C_1v80 (the single corner ORFS optimised at)	—
Package assembled	2026-10-06; design inputs copied from the repository tree at commit 3dffffbb (branch claudе/hopeful-rubin-0yf8io)	Every later commit on the branch changes only review/ or the repository landing page README.md (added after packaging; not a design input), or is empty (git diff --name-only 3dffffbb..HEAD outside review/ lists only README.md). Re-runs made for this package are dated 2026-10-06

Other implementations in the repo (sky130hd baseline at 7.0 ns, variant base; IHP SG13G2) are reference runs only and are NOT the reviewed layout.

Note on commit dates: the commit identifiers above are on the current branch and have the same file trees as the commits the package was built from ([COMMIT_MAP.tsv](#) lists them with tree hashes). All dates in this package come from the tool logs and run records, which are dated 2026-09-29 to 2026-10-06, except times marked "originally committed", which are the commit times of the commits the package was built from (same period). The author and committer dates in the branch's commit metadata do not match these and should not be read as the development timeline (D-54).

Notes on this table, checked for this document:

- MANIFEST.sha256, cited in the RTL row, is at the package root: [MANIFEST.sha256](#), generated last on 2026-10-06 (see [README.md](#) section Manifest; discrepancy register (09_review_notes) D-53). The four RTL sha256 values recomputed at 3dfffbb equal those in [fault/summary.md](#) lines 16-19 and in [04_digital_design/README.md](#); e.g. rtl/orbit_demo.v a3fffb229df3e166694d9259cee45411537118aa2673f84bb26b0f4108f3183a.

Purpose and scope

orbit_demo is the "BUILT / DIGITAL DEMONSTRATOR" box of the ORBIT-AI v0.1 concept brief, not the 16-tile chip ([SPEC.md](#) lines 3-7; [brief](#) p.1). It is a standard-cell digital block: four independent signed INT8 multiply / INT32 accumulate lanes behind a valid/ready input stream, a one-result output buffer, duplicated accumulator and result storage with a sticky mismatch fault-stop, and a three-copy (TMR) thermal state machine that throttles, stops and recovers input admission from a digital temperature input ([orbit_demo.v](#) lines 1-16). It was taken through OpenROAD-flow-scripts on sky130hd to a routed GDS and was not fabricated (brief p.1 "No physical chip fabricated."; `finish_design_instance_count_padcells 0` in [6_report.json](#) line 60). There are no analog blocks, no pad ring and no package. [SPEC.md](#) lines 9-10: "Nothing here is radiation-qualified, timing-signed-off for flight or thermally validated. Thresholds are illustrative."

Item	Demonstrator orbit_demo (built, reviewed here)	ORBIT-AI concept chip (brief only, not built)
Compute	4 lanes x 1 INT8 MAC, INT32 accumulate	16 tiles x 32 x 32 = 16384 INT8 MACs: TARGET
Clock	no requirement (MISSING); 7.2 ns met at TT (ORFS optimised at TT only): VERIFIED; re-time of the same layout passes at ff_n40C_1v95 and fails setup at ss_100C_1v60 (WNS -6.131 ns): REPRODUCED 2026-10-06	200-800 MHz, 6.55-26.21 TOPS: TARGET ("No timing measurement supports these clock targets", p.2)
Storage	521 flip-flops; no SRAM or ECC (rtl/ecc/ exists but is not instantiated: N/A here)	32 MiB SRAM with SECDED 64+8: TARGET
Protection	duplicate + compare fault-stop (512 bits), TMR + repair (6 bits), 3 unprotected flip-flops, no replay	duplication, checkpoint/retry, tile quarantine: TARGET
Power	no budget (MISSING); 52.1092 mW default-activity estimate: VERIFIED tool output	40 W "for conceptual sizing only": ASSUMPTION
Process	SKY130 / sky130_fd_sc_hd via ORFS: VERIFIED	"process, package ... remain undecided" (p.2)

Requirements and verification status

Evidence keys (location and test conditions):

Key	Evidence	Conditions	Status
E-FV	formal_summary.md	make formal, 12 SymbiYosys tasks: k-induction k = 1-8, abc pdr, suprove liveness, cover depth 24/8; Yosys 0.69+154, SBY v0.69, Yices 2.7.0, bitwuzla 0.9.1; reference model, inputs free after reset in cycle 0. datapath_pdr re-run 2026-10-06: PASS, 941 s (make_formal-extra.log , summary_extra.md); published result in formal/summary.md	REPRODUCED 2026-10-06

Key	Evidence	Conditions	Status
E-VAC	formal_vacuity.md	19 one-line RTL mutants, 19 CAUGHT	REPRODUCED 2026-10-06
E-SIM	directed_icarus.log , directed_verilator.log	tb/tb_orbit_demo.v; Icarus 14.0 (devel) and Verilator 5.053; seed 1; zero-delay RTL, 10 ns sim clock; 14 scenarios, 133601 cycles, 0 errors	REPRODUCED 2026-10-06
E-RND	random_icarus_summary.md	cocotb; seeds 1-5 x 20000 cycles + wrap_soak (149226 cycles); 7544 results, 0 mismatches	REPRODUCED 2026-10-06
E-WAV	waveforms/README.txt , negctl_summary.txt	5 self-checking VCD scenarios (Icarus 14.0, RTL zero-delay), all PASS; each FAILs on an RTL copy with its targeted bug	REPRODUCED 2026-10-06
E-GLSF	gls.log	pd/gls/tb_gls.v: RTL and routed sep 6_final.v in lockstep, zero-delay, 30000 cycles, seed 1, one flip per 32 cycles into the 518 redundant bits; run 2026-09-29	VERIFIED
E-FI	fault/summary.md , fault/logs/ ; re-run formal_results.md , make_fault.log	generated RTL copies, single upset per trace (any lane/bit/cycle); prove k-induction $k = 4$, negatives BMC depth 12; SBY v0.69, Yices 2.7.0. Re-run 2026-10-06 (19 tasks, 0 unexpected), in 07_verification/rerun_2026-10-06/fault/	REPRODUCED 2026-10-06
E-SEU	fault/campaign.md ; re-run campaign_report.md (byte-identical)	fault/tb_seu_campaign.v, Icarus 14.0, seed=1 tpb=12 tpb_unprot=200, 6836 trials, flips at the falling edge	REPRODUCED 2026-10-06
E-AUD	storage_audit.txt ; synth/storage_audit.txt	pd/audit_storage.py on sep 6_final.v (18/18 with the older script; current script 19/19, re-run 2026-10-06: storage_audit_current_script.txt); scripts/storage_audit.py on the generic Yosys 0.69+154 netlist	VERIFIED
E-MUT	mutation/summary.md	40 RTL mutants x 5 make targets; evidence bench tb_mutation_gaps.v	VERIFIED
E-PIN	pinout.csv , rtl_ports.json	parse_pins.py on sep 6_final.def; Yosys 0.69+154 elaboration	REPRODUCED 2026-10-06
E-STA	sta_corners_sep_2026-10-06/	pd/sta_corners.tcl on sep 6_final.{odb,sdc,spef}, 7.2 ns, nominal OpenRCX SPEF at every corner	REPRODUCED 2026-10-06

SPEC sections 2-9 (the §9 claim number is given in brackets), then the brief's claims and targets:

Requirement	Source	Status	Evidence
R2 18 ports, names/directions/widths per §2 (80 in, 135 out bits)	SPEC 25-45	REPRODUCED 2026-10-06	E-PIN: RTL and DEF both 18 ports / 215 signal bits, 0 mismatches (see Interfaces)
R3.1 exact signed 8 x 8 product, sign-extended to 32 bits [9.1]	SPEC 52-57	REPRODUCED 2026-10-06	E-FV P1 (product $k = 1$ over all 2^{16} operand pairs; datapath $k = 6$); E-SIM signed_edges; E-VAC zext_product
R3.2 accumulate mod 2^{32} , no saturation or overflow flag [9.5]	SPEC 56	REPRODUCED 2026-10-06	E-FV P1, wrap covers C20-C22; E-SIM int32_wrap; E-RND wrap_soak
R3.3 in_last -> result, out_valid next cycle; in_first && in_last legal; sums continue across in_last; lanes independent	SPEC 59-64	REPRODUCED 2026-10-06	E-FV P1, P2; E-SIM first_and_last, continue_after_last; E-WAV a_mac3
R4.1 §4 out_valid/in_ready equations; full buffer blocks every beat [9.2]	SPEC 68-74	REPRODUCED 2026-10-06	RTL orbit_demo.v 64, 70-71 identical to SPEC; E-FV P2, P4; E-VAC no_backpressure; E-SIM backpressure; E-WAV b_backpressure
R4.2 in_last with out_fire in one cycle replaces the result, no bubble	SPEC 75-76	REPRODUCED 2026-10-06	E-FV P2 (incl. live); E-SIM back_to_back
R4.3 out_data stable while out_valid && !out_ready (no-fault case)	SPEC 77	REPRODUCED 2026-10-06	E-FV P3. SPEC omits that reset/clear_fault withdraw a presented result (see discrepancy register (09_review_notes))

Requirement	Source	Status	Evidence
R4.4 out_ready -> in_ready combinational	SPEC 78	REPRODUCED 2026-10-06	RTL 70-71; timed in-to-out, TT slack 2.577 ns (supplement). clear_fault -> in_ready is also combinational and undocumented (see discrepancy register (09_review_notes))
R5.1 two copies each of acc and result per lane, own feedback, shared product; out_data from copy A	SPEC 82-84	VERIFIED	RTL orbit_mac_lane.v 25-60; E-AUD. Copy-A source checked only by tb_mutation_gaps.v T2 (E-MUT: mutant m07 survives all make targets)
R5.2 mismatch blocks out_valid/in_ready in the same cycle; fault from the next cycle [9.3]	SPEC 86-92	REPRODUCED 2026-10-06	E-FI acc_a/acc_b/res_a/res_b prove PASS; E-SEU 6136 of 6136 detected at latency 1; E-WAV c_fault
R5.3 fault sticky until clear/reset	SPEC 88, 92	REPRODUCED 2026-10-06	E-WAV c2_fault_sticky and its negative control. No make target checks it (E-MUT: m22 fault_not_sticky survives)
R5.4 one acc/result upset never transfers a wrong out_data; it stops	SPEC 93-94	REPRODUCED 2026-10-06	E-FI D_out_fire_matches_ref proven; E-SEU 0 escapes in 6144 trials; E-GLSF 702 of 702 flips stopped on the routed netlist
R5.5 clear_fault/reset clear fault_q, out_valid_q, zero both copies, keep thermal state; no replay	SPEC 95-98	REPRODUCED 2026-10-06	E-FV P8 (clear from an unconstrained state); E-VAC clear_keeps_fault, clear_keeps_acc; E-SIM clear_fault, reset_midrun
R6.1 copies u_thermal.u_copy0/1/2, vote, rewrite every cycle, reset STOP; \$6 next-state table, signed t [9.4]	SPEC 102-116	REPRODUCED 2026-10-06	E-FV P5; E-VAC threshold mutants; E-SIM throttle, stop_and_recover, negative_temps; E-WAV d_thermal (383 checks)
R6.2 admission: NORMAL always, THROTTLE alternate (first admits), STOP never; drain in STOP	SPEC 118-124	REPRODUCED 2026-10-06	E-FV P6, P4; E-SIM throttle, drain_in_stop. SPEC phase wording differs from orbit_thermal_tmr.v line 77 (see discrepancy register (09_review_notes))
R6.3 voted code 3 behaves as STOP	SPEC 43, 115, 124	VERIFIED	only tb_mutation_gaps.v T3 (E-MUT); formal code-3 branches are vacuous; mutant m27 survives every make target
R6.4 a single upset thermal copy is outvoted and repaired; therm_repair	SPEC 126-127	REPRODUCED 2026-10-06	E-FI therm_c0/c1/c2 PASS; E-SEU 72 of 72 REPAIRED; E-GLSF 234 of 234 repaired
R7 521 flip-flop bits, 518 in 9 groups, 3 unprotected; synthesis keeps copies distinct [9.6]	SPEC 129-145, 174	VERIFIED	E-AUD: routed 521 sky130_fd_sc_hd_dfxt1, "RESULT: PASS (18/18 checks passed)" (19/19 with the current script); generic 9/9; without keep_hierarchy copies merge (521 -> 517 bits, audit FAIL)
R8.1 measure out_valid_q escapes	SPEC 156	REPRODUCED 2026-10-06	E-SEU 200 of 200 escape (38 LOST, 162 EXTRA); E-FI neg_out_valid_q FAIL as expected
R8.2 measure fault_q 1->0	SPEC 157-158	MISSING	single-upset flows cannot reach it; needed: a double-upset scenario
R8.3 measure phase	SPEC 159	REPRODUCED 2026-10-06	E-SEU 156 MASKED, 44 TIMING; shifted vs doubled admission not separated
R8.4 measure shared multiplier	SPEC 160	REPRODUCED 2026-10-06	E-FI neg_product: wrong out_data with fault = 0
R8.5 measure comparators, voter, clock, reset, sensor/handshake inputs, common-mode	SPEC 160-162	MISSING	not injected in any flow; needed: transient injection on vote/next-state, in_fire, clr, mismatch, clock, reset
R9.7 faults only in generated copies; no fault ports in production RTL	SPEC 175-176	REPRODUCED 2026-10-06	E-PIN: 18 functional ports, no test port
B1 "521 flip-flop bits"; "Nine storage groups audited"	brief p.4	VERIFIED	R7

Requirement	Source	Status	Evidence
B2 "4,356 generic logic cells"	brief p.4	CLAIM (unverified)	not reproduced: Yosys 0.69+154 gives 4373, YoWASP 0.69 gives 4384 (synth/summary.md lines 12, 21); no script given (see discrepancy register (09_review_notes))
B3 bounded results: "24 bounded SAT time steps", fault "8 time steps", wrap "6 time steps"	brief p.4	CLAIM (unverified)	no script or log; the repo's unbounded proofs (R3-R6) cover the behaviour
B4 "No PDK mapping or physical area estimate"; "OpenROAD was not run"	brief p.4-5	N/A	earlier project state, superseded by this layout (see discrepancy register (09_review_notes))
B5 thresholds 80 / 95 / 70 C, "not device ratings"	brief p.3	TARGET	RTL parameters orbit_demo.v 22-24; behaviour in R6.1
B6 "Separate redundant state physically"	brief p.3	VERIFIED	separation.md : acc/res same-bit centre min 251.17 um, copy A-B flip-flop gap 242.94 um, thermal 97.92 um / gap 92.54 um, 0 of 262 pairs touching. Geometry only. The 20 um / 10 um acceptance limits are a TARGET with no stated rationale
B7 "protect clock/reset distribution"	brief p.3	MISSING	not implemented: one CTS tree for all 521 sinks (4_1_cts.log line 17)
B8 16 tiles x 32 x 32 MACs, 6.55-26.21 TOPS at 200-800 MHz	brief p.1-2	TARGET	concept chip, not built
B9 40 W per chip	brief p.1-2	ASSUMPTION	"not a power estimate" (p.2)
B10 32 MiB SECDED SRAM; 16 GiB+ external memory	brief p.1-2	TARGET	concept chip; N/A for orbit_demo (no SRAM)
B11 "No physical chip fabricated"; sensors, calibration, shutdown circuit not implemented	brief p.1, p.3	N/A	scope statements; consistent with padcells 0 and the digital temp_c input

Manufacturing process

Item	Value	Source	Status
Process	SkyWater SKY130 open PDK, ORFS platform sky130hd	config.mk line 11	VERIFIED
Standard cells	sky130_fd_sc_hd (521 x sky130_fd_sc_hd__dfxtp_1)	synth_stat.txt line 219	VERIFIED
Liberty for synthesis, optimisation, ORFS final STA (TT only), power	sky130_fd_sc_hd__tt_025C_1v80 (1.8 V, 25 C, process 1.0; sha256 ec0e1067a35c8bf2...)	6_report.log line 6; 1_2_yosys.log line 213	VERIFIED
Liberty for informational STA	ss_100C_1v60, ff_n40C_1v95 from the master branch of efabless/skywater-pdk-libs-sky130_fd_sc_hd (16-hex sha256 prefixes only, no commit)	sta_corners_sep.txt lines 8-9	REPRODUCED 2026-10-06
PDK version	not recorded in any log; the TT liberty header has only revision : 1.0000000000. Needed: open_pdk/skywater-pdk commit per run	—	MISSING
Parasitics	OpenRCX, rcx_patterns.rules, one extraction corner; no RC corners	6_report.log lines 16-18	VERIFIED
Tool versions	OpenROAD "unknown"; Yosys 0.68+post (ORFS synthesis); KLayout 0.30.12	6_report.log line 1; 1_2_yosys.log line 1; 6_lvs.log line 1	VERIFIED
ORFS image	run used tag openroad/orfs:latest, no digest logged; requirements-tools.txt line 13 pins sha256:2e5bf6fe865e..., the only local image	—	ASSUMPTION

Item	Value	Source	Status
ORFS/OpenROAD commit, kepler-formal version	not recorded	—	MISSING

Supply voltages

No supply specification exists: SPEC.md, the RTL and the brief state no core or I/O supply voltage, tolerance, temperature range or power-up sequence (MISSING; needed: a SPEC section with VDD nominal/min/max and junction temperature range). The only evidence:

Item	Value	Source	Status
Library corner	TT, 1.80 V, 25 C. A characterisation corner, not a requirement	6_report.log line 6	VERIFIED
Supply nets in the layout	one pair. DEF PINS 217 = 215 signal + VDD (USE POWER) + VSS (USE GROUND), met5 strap pins; SPECIALNETS 2: VDD to VPB/VPWR, VSS to VNB/VGND	6_final.def.gz , decompressed lines 18346-18347, 18362, 19239-19240, 23252	VERIFIED
Static IR analysis supply	"Supply voltage : 1.80e+00 V". Not set in pd/sky130hd_sep/config.mk; it is the platform default PWR_NETS_VOLTAGES ?= VDD 1.8 (image platforms/sky130hd/config.mk line 145, read 2026-10-06)	6_report.log line 32	VERIFIED
Informational STA voltages	1.60 V (SS), 1.95 V (FF)	E-STA	REPRODUCED 2026-10-06
Power intent (UPF/CPF)	none; single domain in the layout. Needed: a SPEC statement that one domain is intended	—	MISSING

Interfaces

SPEC section 2 checked against the RTL ([orbit_demo.v](#) lines 26-51; [rtl_ports.json](#)) and the routed DEF pins ([pinout.csv](#)). Status of the table: REPRODUCED 2026-10-06 (Yosys 0.69+154 elaboration, parse_pins.py). All 18 ports match in name, direction and width.

Port	Dir	Width	Meaning (SPEC §2)	RTL line	DEF pins (edge)
clk	in	1	Clock	26	W
rst_n	in	1	Synchronous reset, active low	27	W
in_valid / in_ready	in / out	1 / 1	Input beat valid / accepted this cycle if in_valid	30 / 31	W / W
in_first / in_last	in	1 / 1	Beat starts a new sum / after this beat copy all sums to the buffer	32 / 33	S / W
in_a, in_b	in	32 each	Signed INT8 operands, lane i at [8*i +: 8]	34, 35	in_a S8 N10 W6 E8; in_b S9 N10 W6 E7
out_valid / out_ready	out / in	1 / 1	Buffer holds a presentable result / consumer takes it	38 / 39	W / W
out_data	out	128	Result, lane i at [32*i +: 32], signed INT32 (copy A)	40	W82 S24 N22
temp_valid	in	1	Thermal reading valid	43	E
temp_c	in	8	Signed whole degrees C (1)	44	E
clear_fault	in	1	Clear fault, zero lane storage, empty buffer	47	E
fault	out	1	Sticky mismatch flag, 1-cycle latency	48	W
therm_state	out	2	0 NORMAL, 1 THROTTLE, 2 STOP, 3 as STOP	49	E
therm_repair	out	1	Thermal copies disagree this cycle	50	E

Port	Dir	Width	Meaning (SPEC §2)	RTL line	DEF pins (edge)
shutdown_req	out	1	therm_state[1]	51	E

(1) Declared input wire [7:0] and reinterpreted as wire signed [7:0] t in [orbit_thermal_tmr.v](#) line 42, so comparisons are signed.

- 215 signal pins: 131 on met3 (E/W edges), 84 on met2 (N/S edges), minimum-size block-edge pins placed by `place_pins`; not a specified pinout, not bond pads. VDD/VSS are not in the SPEC table. Details: 08_pinout_packaging.
- Combinational input-to-output paths: `out_ready` -> `in_ready` (documented) and `clear_fault` -> `in_ready` (undocumented; see discrepancy register (09_review_notes)).
- I/O timing: 1.44 ns (0.2 x period) input and output delay "for the (unknown) outside world" ([constraint.sdc](#) lines 5-7, 20-27): ASSUMPTION. No `set_driving_cell`, `set_load`, `set_input_transition` or `set_clock_uncertainty` in [6_final.sdc](#): MISSING (needs an interface timing budget and clock jitter spec).
- Reset protocol (minimum `rst_n` low time; no on-chip synchronizer) and sensor interface timing (update rate, synchronisation to `clk`): MISSING from SPEC.

Target speed

No speed requirement exists for the demonstrator (MISSING; needed: target frequency and PVT range). The 7.2 ns period ([constraint.sdc](#) line 19) is the shortest period tried that closed at TT with the fences: 7.10 ns fails with setup WNS -0.094 ns ([period_exploration.md](#)). Conditions for all rows: sep layout, 7.2000 ns (138.9 MHz), 1.44 ns I/O delay, propagated clock, nominal OpenRCX SPEF at every corner, no OCV/derate or clock uncertainty, ORFS optimisation at TT only, OpenSTA in OpenROAD "unknown".

Result	Setup WNS / TNS (ns)	Hold WNS (ns)	Min period / fmax	Source	Status
ORFS final STA, TT 25 C 1.80 V	+0.031 / 0	+0.436	7.17 ns / 139.49 MHz	6_finish.rpt lines 5, 15, 20; 6_report.json lines 32-35	VERIFIED
Re-run TT tt_025C_1v80	+0.031 / 0.000	+0.436	7.17 ns / 139.49 MHz	sta_tt.log	REPRODUCED 2026-10-06
Re-run SS ss_100C_1v60	-6.131 / -2463.532, FAILS (516 endpoints)	+0.891	13.33 ns / 75.01 MHz	sta_ss_100C_1v60.log lines 6-11	REPRODUCED 2026-10-06
Re-run FF ff_n40C_1v95	+2.276 / 0.000	+0.281	4.92 ns / 203.07 MHz	sta_ff_n40C_1v95.log	REPRODUCED 2026-10-06

- TT worst setup path `in_a[27] -> g_lane[3].u_lane.u_res_a/q[29]$_SDFFE_PN0P_` includes the assumed 1.44 ns input delay; 139.49 MHz is from `report_clock_min_period -include_port_paths`. TT reg-to-reg worst slack is 1.346 ns ([supplement](#)).
- SS fails also reg-to-reg (-4.420 ns) and has 37 max-slew and 12 max-capacitance violating pins; TT and FF have none.
- Brief 200-800 MHz: TARGET for the concept chip only; 200 MHz is above the demonstrator's TT fmax.
- Throughput: no tool reports it. `docs/research/space-compute-competitors.md` line 25 "1.1 GOPS INT8 peak (4 MACs x 2 ops x 139 MHz)": CLAIM (unverified), a TT-only arithmetic bound.
- RC corners, other PVT corners, SDF-annotated simulation and multi-corner optimisation: MISSING.

Power budget

No power budget exists for the demonstrator (MISSING; needed: a target from the design owner). The brief's 40 W per chip is a concept-chip sizing ASSUMPTION and does not apply to `orbit_demo`.

Result	Value	Conditions / source	Status
Total power, routed sep layout	52.1092 mW: internal 26.3633 mW, switching 25.7459 mW, leakage 0.0279 uW	OpenSTA report_power (OpenROAD "unknown"), 7.2 ns, TT 25 C 1.80 V, OpenRCX SPEF, default activity; 6_report.json line 49, results.md lines 53-56	VERIFIED (tool estimate, not a measurement)
By group	Sequential 5.96 mW (11.4 %), Combinational 43.2 mW (83.0 %), Clock 2.90 mW (5.6 %), Macro 0, Pad 0	power_default_activity.txt ; 6_finish.rpt lines 549-560	VERIFIED
Activity assumed	the files state only "the tool's DEFAULT switching activity (no simulation activity annotated)". The ORFS scripts in the image call report_power with no set_power_activity, read_vcd or read_saif (grep of flow/scripts, 2026-10-06), so OpenSTA's built-in default applies; no log prints its toggle rate or static probability. Needed: VCD/SAIF from a representative workload read into OpenSTA	power_default_activity.txt	MISSING
Static IR drop, same activity	VDD 0.4539 mV, VSS 0.6275 mV worst (0.03 %)	PSM, 1.80 V; 6_report.log lines 27-49	VERIFIED
Power at SS/FF, dynamic IR, electromigration	not run	—	MISSING

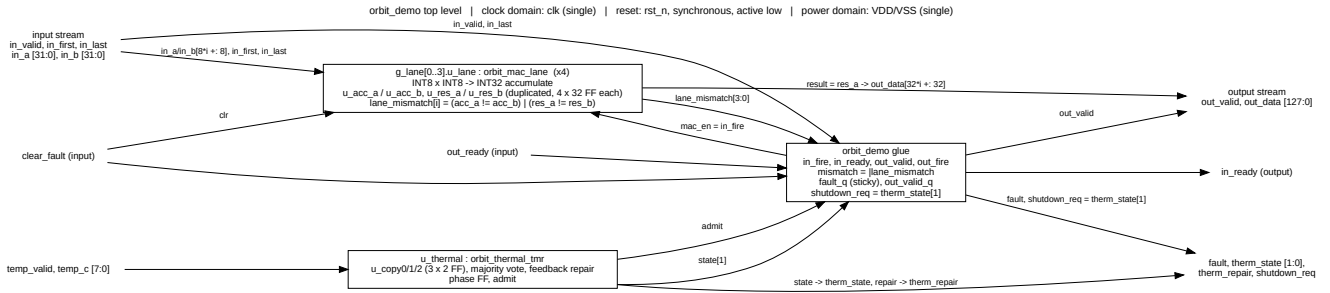
Area and cell counts

Conditions: ORFS sky130hd sep run 2026-09-29, TT liberty; ORFS synthesis by Yosys 0.68+post with abc -D 7.2.

Quantity	Value	Source	Status
Die	346.295 x 346.295 um = 119920 um ²	2_1_floorplan.log line 20; DEF DIEAREA	VERIFIED
Core	(2.300, 2.720)-(344.080, 342.720) um = 341.780 x 340.000 um = 116205.2 um ² ; 125 rows x 743 unithd sites	2_1_floorplan.log lines 19-22	VERIFIED
Utilisation	CORE_UTILIZATION 40 % (config); 0.403 at floorplan; 0.536 final (cell area excl. fill / core)	config.mk line 35; 2_1_floorplan.log line 24; 6_report.json line 64	VERIFIED
Cell area	62299.8 um ² excl. fill incl. tap; 60392.97 um ² excl. fill and tap	6_report.log line 67; results.md lines 42-43	VERIFIED
Instances	18175: fill 10381, tap 1524, antenna 20, clock buffer 88, clock inverter 8, timing-repair buffer 1201, inverter 171, sequential 521, multi-input combinational 4261. 6270 excl. fill/tap; 6_final.v has 6250 leaf cells	6_report.log lines 51-61; storage_audit.txt line 3	VERIFIED
Macros, pad cells	0, 0	6_report.json	VERIFIED
Synthesised netlist	5130 cells, 46866.1984 um ² (sequential 10430.0032 um ² , 22.25 %); 19 kept orbit_keep_reg instances	synth_stat.txt lines 72, 268-269	VERIFIED
Generic synthesis	4899 cells = 4373 logic + 521 flip-flops + 5 \$scopeinfo (Yosys 0.69+154, no area)	synth/synth_counts.txt	VERIFIED

Only the flip-flop count (521) is the same across the generic, synthesised and routed netlists; the other counts are not comparable. Layout views: 06_physical_design.

Top-level block diagram



Drawn from rtl/*.v at 26d71a8 (Graphviz source [orbit_demo_top_level.dot](#); [PDF](#)). Instance names are the RTL names. One clock domain (clk), one synchronous active-low reset (rst_n), one supply pair (VDD/VSS). Detailed diagram: [orbit_demo_block_diagram.svg](#).